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Sun Exposure and Photoprotection Knowledge, Habits, and Attitudes Among Spanish Military Personnel

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ABSTRACT

Background and Objectives: The incidence of skin cancer continues to rise, with ultraviolet (UV) radiation exposure representing its main modifiable risk factor. Military personnel constitute a particularly vulnerable population due to the combination of occupational and recreational sun exposure, together with operational constraints related to uniforms and working conditions. The aim of this study was to assess knowledge, habits, and attitudes related to sun exposure and photoprotection among Spanish military personnel, and to explore differences in photoprotective practices according to sociodemographic and professional variables.

Materials and Methods: A cross-sectional observational study was conducted using a validated self-administered questionnaire in a sample of 500 active-duty Spanish military personnel from different ranks. Data collected included sociodemographic and phenotypic characteristics, knowledge, habits, and attitudes. Descriptive analyses and bivariate inferential analyses using non-parametric tests were performed.

Results: Most participants reported frequent sun exposure in both occupational and leisure settings, with longer daily exposure during leisure time; 81.6% reported exposure of ≥ 3 h per day. Overall, 73.4% experienced at least one sunburn in the previous year. Sunglasses and sunscreen were the most commonly adopted photoprotective measures, whereas avoidance of peak sun hours and use of protective clothing were infrequent. Knowledge of UV radiation risks was generally high, although misconceptions persisted regarding shade protection and vitamin D exposure. Photoprotective practices were significantly poorer during leisure time than during occupational activities ($p < 0.01$).

Conclusions: Spanish military personnel demonstrate adequate knowledge and positive attitudes toward photoprotection but insufficient protective practices, particularly outside the occupational setting. These findings highlight the need to strengthen health education and integrate photoprotection into military training and operational environments.

1 | Introduction

The incidence of skin cancer has increased steadily over recent decades, becoming one of the most prevalent malignancies in fair-skinned populations [1]. This upward trend has been consistently documented in Europe and Spain, particularly for non-melanoma skin cancers, and is largely attributed to cumulative ultraviolet radiation (UVR) exposure and population

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aging. Recent epidemiological estimates confirm skin cancer as a growing public health concern, with a sustained rise in incidence over the last two decades [2].

Solar UVR is the principal environmental risk factor in the development of both melanoma and non-melanoma skin cancers [3]. UVR induces direct DNA damage, promotes photoaging and contributes to local and systemic immunosuppression, thereby facilitating carcinogenesis [4]. Two main patterns of sun exposure have been described: Chronic exposure, typically occupational in nature, and intermittent exposure, more commonly associated with recreational activities [5]. While both contribute to cumulative actinic damage, chronic exposure has been more strongly associated with keratinocyte carcinomas, whereas intermittent and intense exposure plays a more prominent role in melanoma development.

Outdoor workers constitute a population at increased risk due to prolonged and repeated exposure to solar UVR during working hours. This group includes agricultural workers, construction workers, lifeguards, athletes and military personnel, among others [5, 6]. Occupational exposure is often exacerbated by limited access to shade, rigid schedules and insufficient implementation of effective photoprotective measures. Epidemiological studies have shown that outdoor workers experience significantly higher cumulative UVR doses and present higher rates of actinic keratoses and non-melanoma skin cancer compared with indoor workers [6].

Military personnel represent a particularly vulnerable subgroup within outdoor workers [7]. Their professional activities frequently involve prolonged sun exposure during physical training, maneuvers, and operational deployments, often under extreme environmental conditions and with limited opportunity to modify exposure patterns [7, 8]. Uniform regulations and operational constraints may further hinder the consistent use of physical and topical sun protection [8]. A systematic review conducted in military populations has reported an increased incidence of both melanoma and non-melanoma skin cancers compared with the general population, highlighting the relevance of occupational UVR exposure in this group [7].

Despite this elevated risk, adherence to photoprotective behaviors among military personnel remains suboptimal. Studies carried out in military and veteran populations have described high frequencies of sunburns, inadequate sunscreen use, and permissive attitudes toward tanning, even among individuals with sufficient knowledge of sun-related risks [8, 9]. This dissociation between knowledge and behavior has also been observed in other outdoor-exposed populations, including athletes and outdoor workers [10].

In Spain, research on sun exposure and photoprotection has primarily focused on civilian populations and outdoor athletes, such as runners and other sports practitioners. Among the sports that have been specifically studied are skating [11], golf [12], aquatic sports [13], beach sports [14], and mountain sports [6]. These studies consistently report high levels of sun exposure, frequent sunburns and insufficient use of protective clothing and sunscreen, despite relatively good awareness of skin cancer risks [14–16]. However, evidence specifically addressing sun exposure habits, photoprotection practices, and attitudes among Spanish military personnel remains scarce.

Given the occupational characteristics of military service and the lack of specific national data, further research is needed to better understand sun exposure patterns and photoprotective attitudes in this population. Identifying modifiable risk factors across different military contexts may contribute to the development of targeted prevention strategies aimed at reducing cumulative UVR exposure and long-term skin cancer risk.

Therefore, the aim of the present study was to evaluate knowledge, habits, and attitudes related to sun exposure and photoprotection among Spanish military personnel across three settings: physical training and sports activities, maneuvers and missions, and outdoor leisure time, in order to identify opportunities for targeted preventive interventions within the Armed Forces.

2 | Materials and Methods

2.1 | Participants

A descriptive cross-sectional study was conducted between September and November 2025 in a sample of 500 active-duty Spanish military personnel from different ranks (enlisted personnel, non-commissioned officers, and officers). A non-probabilistic convenience sampling method was used, including those who voluntarily and anonymously agreed to participate after receiving information about the study.

Inclusion criteria were active-duty military status, residence within Spanish national territory, and provision of written informed consent. Exclusion criteria included refusal or failure to provide informed consent and questionnaires with more than 10% missing responses, which were excluded from the analysis.

2.2 | Questionnaire

Data were collected using a structured questionnaire based on the habits, attitudes, knowledge and sun exposure questionnaire (CHACES) [17], previously validated in Spanish populations. Additional items were included to collect military and professional information (rank, corps, and years of service) and to assess sun exposure habits and photoprotective behaviors across three differentiated contexts: Physical training or sports activities, operational activities (missions, maneuvers, or navigation), and outdoor leisure time.

The final questionnaire consisted of ten sections covering: Sociodemographic and professional data; phenotypic characteristics and skin phototype; personal and family history of premalignant lesions and skin cancer; sun exposure habits in different contexts; sunburn episodes in the previous year; photoprotection practices; attitudes related to sun exposure; knowledge about the effects of UVR; and aspects related to photoprotective information and culture within the Armed Forces.

Questions included closed-ended items, five-point Likert scales, and dichotomous questions (true/false or yes/no), following the

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original CHACES structure. Estimated completion time was 10–15 min. The questionnaire was self-administered and anonymous, ensuring confidentiality and pseudonymized data processing in accordance with current data protection regulations.

Photoprotective practices were assessed using a specific scale (PPS), composed of Likert-type items that measure the frequency of use of different sun protection measures in three contexts: Physical training/sport, missions or maneuvers, and outdoor leisure. A total score (PPS total) and context-specific subscale scores were calculated. Higher scores indicated better photoprotective practices. Reverse-worded items were recoded beforehand to ensure consistency of scale direction.

2.3 | Statistical Analysis

Data were analysed using IBM SPSS Statistics for Windows, version 29.0 (IBM Corp., Armonk, NY, USA). Qualitative variables were described using absolute frequencies and percentages. Quantitative and ordinal variables were summarized using median and interquartile range (IQR), after confirming non-normal distribution through graphical inspection and the Kolmogorov–Smirnov test.

Bivariate non-parametric tests were used for inferential analysis. Comparisons between two independent groups were performed using the Mann–Whitney *U* test, and comparisons among more than two groups using the Kruskal-Wallis test. Associations between quantitative or ordinal variables were assessed using Spearman's correlation coefficient. Statistical significance was set at $p < 0.05$.

No questionnaires were excluded; therefore, all analyses were conducted on the full sample ($n = 500$).

2.4 | Ethics Committee

The study was approved by the Research and Animal Experimentation Ethics Committee of the University of Alcalá (code CEID/2025/2/065). All participants provided prior informed consent, and data were treated anonymously and confidentially, in accordance with Regulation (EU) 2016/679 and Organic Law 3/2018 on Personal Data Protection and digital rights.

3 | Results

3.1 | Participant Characteristics

A total of 500 active-duty military personnel from different branches and ranks of the Spanish Armed Forces completed the questionnaire and were included in the analysis. 81.4% ($n = 407$) were men and 18.6% ($n = 93$) were women, with a median age of 34 years (IQR: 14; range: 20–55).

Spanish nationality was predominant (99.2%), and most participants belonged to the enlisted ranks (62.8%), followed by officers (20.1%) and non-commissioned officers (17.2%). Regarding marital status, 49.6% were married or living with a partner, 46.2% were single, and 4.2% were separated or divorced. Educational level was mainly secondary education (60.7%), followed by university education (31.4%) and primary education (7.9%). The median length of military service was 11 years (IQR: 15), and 44.8% of participants reported having children under 12 years of age.

With respect to phenotypic characteristics, skin phototypes III-IV predominated (69.0%). During the previous year, 56.4% ($n = 282$) reported having experienced one or two sunburn episodes after sun exposure, 17.0% ($n = 85$) reported three or more episodes, whereas 26.6% ($n = 133$) reported no sunburns (Table 1).

3.2 | Sun Exposure Habits

Regarding sun exposure habits, most participants reported sun exposure for more than 30 days per year, particularly during physical training or sports activities (79.0%), followed by missions, navigation or maneuvers (60.0%), and outdoor leisure or holidays (62.0%). Concerning daily exposure duration, occupational exposure was predominantly 1–2 h per day both during physical training (70.0%) and during missions (41.0%). In contrast, leisure-time exposure showed a more intense and prolonged pattern, with 81.6% of participants reporting exposure of ≥ 3 h per day, indicating a higher-risk recreational profile (Table 2).

3.3 | Photoprotective Practices

With regard to photoprotective practices, sunglasses were the most frequently used photoprotective measure across all three contexts analysed, with frequent or habitual use reported by 74.9% during physical training, 75.2% during missions, and 66.6% during leisure time. The use of sunscreen (SPF > 15) was similar across all settings, with approximately 48%–49% of the participants reporting habitual use. The use of caps or hats was more frequent during occupational activities (53%–54%) than during leisure time (37%). The use of long-sleeved clothing or long trousers was clearly uncommon in all contexts (≤ 10%). Avoidance of sun exposure during peak hours (11:00–17:00) was reported by approximately 40% of respondents, with no relevant differences between settings, while habitual use of shade ranged from 36% to 40% (Table 3).

Inferential analysis showed significant differences in photoprotective practice scores according to sex, with higher values among women in both the total score and the sports and leisure subscales ($p < 0.01$). Significant differences were also observed according to the presence of children under 12 years of age, with higher total and leisure subscale scores among participants with children ($p \leq 0.02$). Age was positively correlated with photoprotective practice scores in the total scale and in the sports, missions and leisure subscales, indicating better practices with increasing age (Spearman correlation, $p < 0.001$). Significant differences were also observed according to military rank, with higher scores among officers and non-commissioned officers compared with enlisted personnel ($p < 0.01$). Likewise, educational level was associated with significant differences in the total scale and in the sports and leisure subscales, with better scores among participants with university education ($p \leq 0.02$). No statistically significant differences in photoprotective practices were observed according to skin phototype (Table 6).

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TABLE 1 | Demographic and professional characteristics of the study population.

Demographic data	*n*	Percentage
Sex		
Male	407	81.4%
Female	93	18.6%
Age		
Median (IQR)	34 (14)	
Nationality		
Spanish	496	99.2%
Foreign	4	0.8%
Educational level		
No education/Primary	40	8.0%
Secondary	304	60.8%
University	156	31.2%
Marital status		
Single	231	46.2%
Married	248	49.6%
Separated/divorced	21	4.2%
Having children under 12 years old		
Yes	224	44.8%
No	276	55.2%
Military rank		
Enlisted personnel/Seamen	314	62.8%
Non-commissioned officers	86	17.2%
Officers	100	20.0%
Years of service in the Armed Forces		
Median (IQR)	11 (15)	
None	133	26.6%
During the previous year, how many times did your skin get sunburned (redness and pain) due to sun exposure?		
1 to 2 times	282	56.4%
3 to 5 times	67	13.4%
6 to 10 times	12	2.4%
More than 10 times	6	1.2%

Note: Data are expressed as absolute frequencies and percentages unless otherwise indicated.
Abbreviation: IQR, interquartile range.

3.4 | Attitudes Toward Sun Exposure and Photoprotection

Regarding attitudes toward sun exposure, most participants showed a favorable attitude toward photoprotection. 91.4% reported concern about sunburn, 92.3% about the risk of skin cancer, and 90.6% considered sunscreen use worthwhile. Nevertheless, permissive attitudes toward tanning persisted: 37.7% reported liking being tanned, and 16.3% acknowledged frequent sunbathing (Table 4).

Inferential analysis showed significant differences according to sex, with more protective attitudes among women ($p = 0.002$), as well as among participants with children under 12 years of age ($p = 0.04$) and those who had experienced recent sunburns ($p < 0.001$). No relevant differences were observed according to military rank or educational level (Table 6).

3.5 | Knowledge About Ultraviolet Radiation

In the knowledge scale, overall knowledge level was high, with an overall proportion of correct answers close to 80%. The best-answered items were those related to the carcinogenic effect and skin aging induced by UVR (97.4%), the importance of childhood photoprotection (89.8%–94.5%), and the need to take extra precautions when the UV index exceeds 3 (92.1%). In contrast, the items with the lowest correct response rates were 'being in the shade eliminates the risk of radiation' (69.4%) and 'it is advisable to sunbathe for at least one hour per day to ensure adequate vitamin D levels' (32.6%), reflecting the persistence of myths and conceptual errors (Table 5).

Participants who had experienced recent sunburns obtained significantly higher scores on the knowledge scale compared with those without a history of sunburn ($p < 0.001$) (Table 6).

3.6 | Institutional Aspects of Photoprotection

Finally, institutional training in photoprotection was limited: only 15.0% of participants reported having received specific information within the Armed Forces. In addition, only 2.0% were aware of the AEDV Piel Sana Foundation, and 8.0% routinely checked the UV index before sun exposure. 73.0% considered that military uniform regulations favour the use of photoprotective measures; however, 70.0% acknowledged using photoprotective measures outside the military context that they do not apply while on duty.

4 | Discussion

This study provides a comprehensive and updated description of sun exposure habits, photoprotective practices, attitudes and knowledge among Spanish military personnel, a population exposed to a combination of occupational and recreational UVR. Beyond confirming high overall exposure, the results highlight a high prevalence of recent sunburns despite predominantly intermediate skin phototypes, a clear dissociation between knowledge and protective behavior, and a limited institutional role in photoprotection training, indicating that cumulative risk is driven not only by unavoidable occupational exposure but also by modifiable behavioral and organizational factors [4, 5, 10].

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TABLE 2 | Sun exposure habits according to context.

	Physical training/sports		Deployments/ missions/maneuvers		Outdoor leisure/ holidays	
Sun exposure habits	***n***	**Percentage**	***n***	**Percentage**	***n***	**Percentage**
Days of sun exposure per year						
Less than 5 days	18	3.6%	59	11.8%	6	1.2%
6–30 days	82	16.4%	140	28.0%	177	35.4%
31–90 days	163	32.6%	165	33.0%	179	35.8%
More than 90 days	237	47.4%	136	27.2%	138	27.6%
Hours of sun exposure per day						
None	30	6.0%	47	9.4%	0	0%
1–2 h	350	70.0%	205	41.0%	92	18.4%
3–4 h	90	18.0%	135	27.0%	208	41.6%
5–6 h	12	2.4%	63	12.6%	128	25.6%
More than 6 h	18	3.6%	50	10.0%	72	14.4%

Note: Sun exposure was self-reported for the previous year. Percentages are calculated by column.

TABLE 3 | Photoprotective practices by exposure context.

	Physical training/sports		Deployments/ missions/maneuvers		Outdoor leisure/ holidays	
Sun protection practices	***n***	**Percentage**	***n***	**Percentage**	***n***	**Percentage**
Use of shade						
Never–rarely–sometimes	300	60.0%	318	63.6%	311	62.2%
Almost always–always	200	40.0%	182	36.4%	189	37.8%
Use of sunglasses						
Never–rarely–sometimes	128	25.6%	127	25.4%	170	34.0%
Almost always–always	372	74.4%	373	74.6%	330	66.0%
Use of hat/cap						
Never–rarely–sometimes	236	47.2%	231	46.2%	315	63.0%
Almost always–always	264	52.8%	269	53.8%	185	37.0%
Wearing long sleeves/long trousers						
Never–rarely–sometimes	461	92.2%	444	88.8%	467	93.4%
Almost always–always	39	7.8%	56	11.2%	33	6.6%
Avoiding midday sun						
Never–rarely–sometimes	303	60.6%	287	57.4%	309	61.8%
Almost always–always	197	39.4%	213	42.6%	191	38.2%
Using sunscreen SPF > 15						
Never–rarely–sometimes	261	52.2%	254	50.8%	254	50.8%
Almost always–always	239	47.8%	246	49.2%	246	49.2%

Note: Values represent the distribution of participants according to frequency of use.
Abbreviation: SPF, sun protection factor.

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TABLE 4 | Attitudes toward sun exposure and photoprotection.

Attitudes	*n*	Percentage
I like sunbathing		
Never–rarely–sometimes	414	82.8%
Almost always–always	86	17.2%
I like being tanned		
Never–rarely–sometimes	309	61.8%
Almost always–always	191	38.2%
The sun makes me feel good		
Never–rarely–sometimes	344	68.8%
Almost always–always	156	31.2%
I don't like using sunscreen		
Never–rarely–sometimes	386	77.2%
Almost always–always	114	22.8%
It's worth using sunscreen		
Never–rarely–sometimes	46	9.2%
Almost always–always	454	90.8%
At midday I prefer being in the shade rather than in the sun		
Never–rarely–sometimes	113	22.6%
Almost always–always	387	77.4%
I worry about getting sunburned		
Never–rarely–sometimes	46	9.2%
Almost always–always	454	90.8%
I worry about spots and wrinkles caused by the sun		
Never–rarely–sometimes	87	17.4%
Almost always–always	413	82.6%
I worry about getting skin cancer from the sun		
Never–rarely–sometimes	39	7.8%
Almost always–always	461	92.2%
It's easy to protect myself from the sun by wearing a hat and protective clothes		
Never–rarely–sometimes	122	24.4%
Almost always–always	378	75.6%

Note: Attitudes were assessed using Likert-type items and grouped into two categories for analysis.

Regarding phenotypic characteristics, skin phototypes III-IV predominated in the present sample (69.0%), which is consistent with the distribution described in Mediterranean populations [18]. Despite this, sunburn prevalence was high: 73.4% of participants reported at least one sunburn episode during the previous year, and 16.4% reported three or more episodes. These figures are comparable to those reported in Spanish endurance athletes and outdoor sports practitioners, where more than half of participants experience recent sunburns [14, 15], and are similar to those observed in military populations from other countries [8]. This finding reinforces that intermediate skin phototypes do not provide effective protection against excessive or repeated UVR exposure [1].

Sun exposure habits revealed a mixed but clearly high-risk profile. Most participants reported exposure on more than 30 days per year during occupational activities, particularly physical training and missions. However, leisure-time exposure was markedly more intense, with 81.6% reporting daily exposure of three or more hours, compared with shorter exposure durations during duty-related activities. This pattern has been consistently described in runners, beach athletes, and other outdoor sports populations, in whom leisure-time exposure is often less regulated and less protected than occupational exposure [14, 15, 19]. In regions with high annual solar irradiance, such cumulative exposure substantially increases lifetime UVR dose and long-term skin cancer risk [6].

Photoprotective practices in this study were characterized by partial and inconsistent adherence. Sunglasses were the most frequently used protective measure, followed by sunscreen and headgear, whereas protective clothing such as long-sleeved garments or long trousers was rarely used ($\leq$10%). Avoidance of peak UVR hours was uncommon, consistent with findings in other European outdoor-exposed groups [20, 21].

Although photoprotective practices were slightly more frequent during occupational activities than during leisure time, overall adherence remained insufficient. This suggests that structured environments may facilitate some degree of protection, whereas leisure contexts favor riskier behaviors, a phenomenon also described in athletes and recreational sports practitioners [19]. From a preventive perspective, leisure-time exposure emerges as a critical and modifiable risk factor that is often underestimated in military populations.

Attitudes toward sun exposure were generally favorable, with more than 90% of participants expressing concern about sunburn and skin cancer. Nevertheless, 37.7% reported liking being tanned, and 16.3% acknowledged frequent sunbathing, reflecting persistent sociocultural beliefs that associate tanned skin with health and attractiveness. This cultural component has been consistently documented in Spanish adolescents, young adults, and broader European populations [18, 22]. In line with previous studies, women and participants with young children showed more protective attitudes, a pattern also observed in medical students and other civilian populations [21, 23, 24].

Overall knowledge regarding UVR and photoprotection was high, with approximately 80% of correct responses, comparable to levels reported in Spanish university students and medical trainees [23] and higher than those observed in some non-European military cohorts [9]. However, important misconceptions persisted, particularly regarding protection in the shade and the need for sun exposure to ensure adequate vitamin D levels. These misconceptions have also been described in population-based studies and athletic groups [18, 22]. The

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TABLE 5 | Knowledge about ultraviolet radiation and photoprotection.

Knowledge	*n*	Percentage
Using tanning beds before the age of 30 increases the risk of melanoma by 75%		
Error	127	25.4%
Correct	373	74.6%
Ultraviolet radiation causes accelerated skin aging and various forms of skin cancer		
Error	13	2.6%
Correct	487	97.4%
When staying in the shade, we are not exposed to the harmful effects of solar radiation		
Error	153	30.6%
Correct	347	69.4%
Using sunscreen is the most effective way to protect oneself from the sun and prevent skin cancer		
Error	71	14.2%
Correct	429	85.8%
Once the skin is tanned, it is no longer necessary to use sunscreen		
Error	27	5.4%
Correct	473	94.6%
Babies under 1 year of age should not be directly exposed to the sun		
Error	50	10.0%
Correct	450	90.0%
It is necessary to take extra sun protection measures when the UV Index is higher than 3		
Error	39	7.8%
Correct	461	92.2%
Dark-colored clothing protects better from the sun than light-colored clothing		
Error	148	29.6%
Correct	352	70.4%
It is advisable to spend at least 1 h in the sun each day to ensure adequate vitamin D levels		
Error	337	67.4%
Correct	163	32.6%
Children should use sunscreen with a protection factor (SPF) equal to or higher than 30		
Error	27	5.4%
Correct	473	94.6%

Note: Percentages represent response distribution.
Abbreviations: SPF, sun protection factor; UV, ultraviolet.

coexistence of adequate knowledge and insufficient protective behavior reinforces the well-documented dissociation between knowledge and practice in skin cancer prevention [4].

Interestingly, participants who reported recent sunburn episodes showed significantly higher levels of knowledge regarding ultraviolet radiation and photoprotection. This finding may reflect a reactive pattern of knowledge acquisition, whereby awareness of sun-related risks increases following the experience of an adverse event. Similar observations have been reported in other outdoor-exposed populations, including athletes and military personnel. However, despite higher knowledge levels, photoprotective practices remained insufficient, reinforcing the well-described dissociation between knowledge and preventive behavior in skin cancer prevention. This finding highlights the limitations of reactive learning and supports the need for proactive, structured preventive interventions implemented before sunburn occurs [4, 9, 10].

In this context, one of the most concerning findings of this study is the limited institutional involvement in photoprotection. Only 15.0% of participants reported having received specific training

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TABLE 6 | Factors associated with photoprotective practices, attitudes, and knowledge (non-parametric analysis).

Variable	Test	Scale/Subscale	Statistic	*p*
Sex	Mann–Whitney *U*	PPS total	$U = 14{,}410$	0.004
		PPS sport	$U = 13{,}185$	< 0.001
		PPS leisure	$U = 14{,}408$	0.003
		Attitudes	$U = 14{,}172$	0.002
		Knowledge	$U = 15{,}766$	0.048
Age	Spearman correlation	PPS total	$r = 0.33$	< 0.001
		PPS sport	$r = 0.27$	< 0.001
		PPS missions	$r = 0.27$	< 0.001
		PPS leisure	$r = 0.31$	< 0.001
		Attitudes	$r = 0.12$	0.005
Military rank	Kruskal–Wallis	PPS total	$\chi^2 = 20.12$ (df = 2)	< 0.001
		PPS sport	$\chi^2 = 17.54$ (df = 2)	< 0.001
		PPS missions	$\chi^2 = 16.55$ (df = 2)	< 0.001
		PPS leisure	$\chi^2 = 12.66$ (df = 2)	0.002
		Knowledge	$\chi^2 = 14.13$ (df = 2)	0.001
Educational level	Kruskal–Wallis	PPS total	$\chi^2 = 7.61$ (df = 2)	0.020
		PPS sport	$\chi^2 = 7.99$ (df = 2)	0.020
		PPS leisure	$\chi^2 = 9.20$ (df = 2)	0.001
		Knowledge	$\chi^2 = 12.07$ (df = 2)	0.002
Marital status	Kruskal–Wallis	PPS total	$\chi^2 = 10.62$ (df = 2)	0.005
		PPS sport	$\chi^2 = 9.07$ (df = 2)	0.010
		PPS missions	$\chi^2 = 5.52$ (df = 2)	0.060
		PPS leisure	$\chi^2 = 10.60$ (df = 2)	0.005
		Knowledge	$\chi^2 = 14.13$ (df = 2)	< 0.001
Children < 12 years	Mann–Whitney *U*	PPS total	$U = 23{,}709$	0.006
		PPS sport	$U = 24{,}422$	0.020
		PPS leisure	$U = 23{,}074$	< 0.001
		Attitudes	$U = 25{,}055$	0.040
Skin phototype	Kruskal–Wallis	PPS leisure	$\chi^2 = 7.05$ (df = 3)	0.070
		Knowledge	$\chi^2 = 7.20$ (df = 3)	0.060
Sunburn episodes	Kruskal–Wallis	Attitudes	$\chi^2 = 16.97$ (df = 3)	< 0.001
		Knowledge	$\chi^2 = 20.71$ (df = 3)	< 0.001

Note: Non-parametric statistical tests were used. Comparisons between two independent groups were performed using the Mann–Whitney *U* test, and comparisons among more than two groups were performed using the Kruskal–Wallis test. Associations between quantitative or ordinal variables were assessed using Spearman's rank correlation coefficient. Photoprotective practices were evaluated using the Photoprotective Practices Score (PPS), including the total score and context-specific subscales. Statistical significance was set at $p < 0.05$.

on sun protection within the Armed Forces, and awareness of national prevention initiatives was minimal. Similar gaps have been identified in other occupational and military settings, where sun protection is often framed as an individual responsibility rather than an institutional priority [7, 9]. Evidence from occupational health research indicates that effective prevention requires the integration of organizational measures, leadership engagement and structural adaptations alongside individual education [5, 18].

Taken together, these findings depict a mixed risk profile characterized by adequate knowledge and generally favorable

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attitudes but insufficient implementation of photoprotective behaviors, particularly during leisure time. Comparable profiles have been described in other outdoor-exposed and uniformed populations, including athletes and outdoor workers, suggesting that preventive strategies should be cross-cutting and adapted to the specific constraints of uniformed professions [14, 20, 21].

Several limitations should be acknowledged. The cross-sectional design precludes causal inference, and the self-reported nature of the data may lead to overestimation of protective behaviors. Nevertheless, the large sample size, inclusion of different military ranks and exposure contexts, and use of a validated questionnaire strengthen the robustness and comparability of the findings [18, 23]. The non-probabilistic convenience sampling may limit generalizability to the entire Armed Forces.

In conclusion, Spanish military personnel exhibit high levels of sun exposure and sunburn prevalence, combined with adequate knowledge and generally favorable attitudes but insufficient adherence to photoprotective practices, particularly outside professional activities. The integration of structured sun education programs into military training, together with organizational measures and adaptations to uniforms and equipment to limit UVR exposure, is essential to embed photoprotection within the preventive culture and operational health framework of the Armed Forces [5, 7].

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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